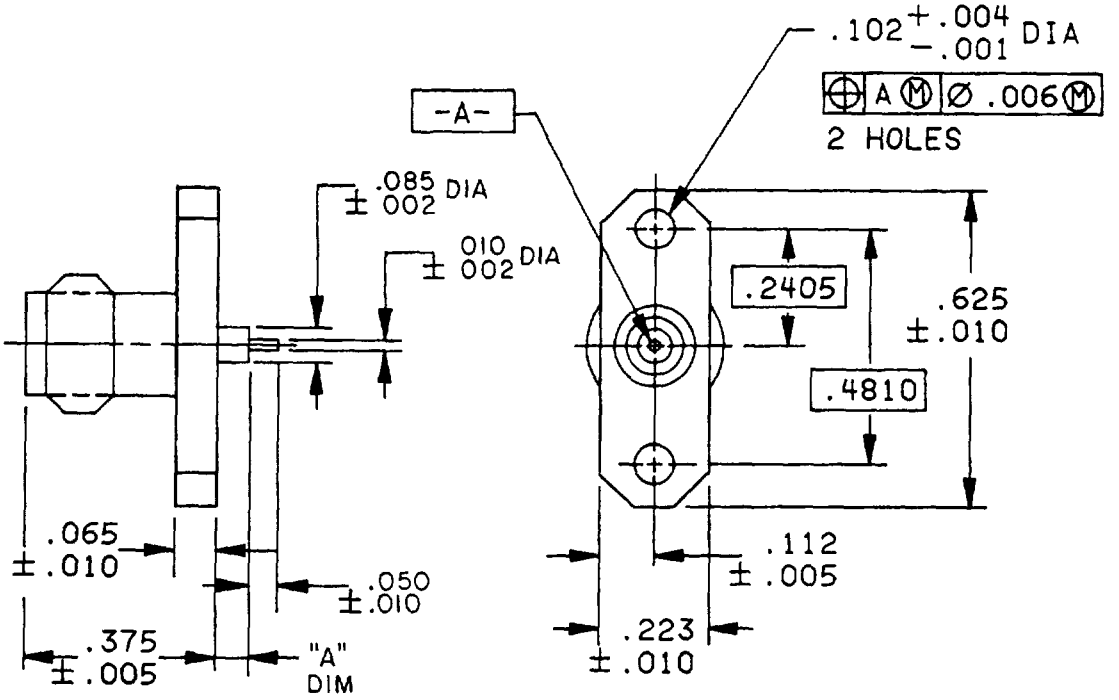
MILITARY SPECIFICATION SHEET

CONNECTOR, COAXIAL, RADIO FREQUENCY, STRIP OR MICROSTRIP TRANSMISSION LINE, SERIES SMA (SOCKET CONTACT, FLANGE MOUNTED EXTENDED DIELECTRIC RECEPTACLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-83517.



PART NO.	"A" DIM
-31001	.057 ±.003
-31002	.125 ± 003

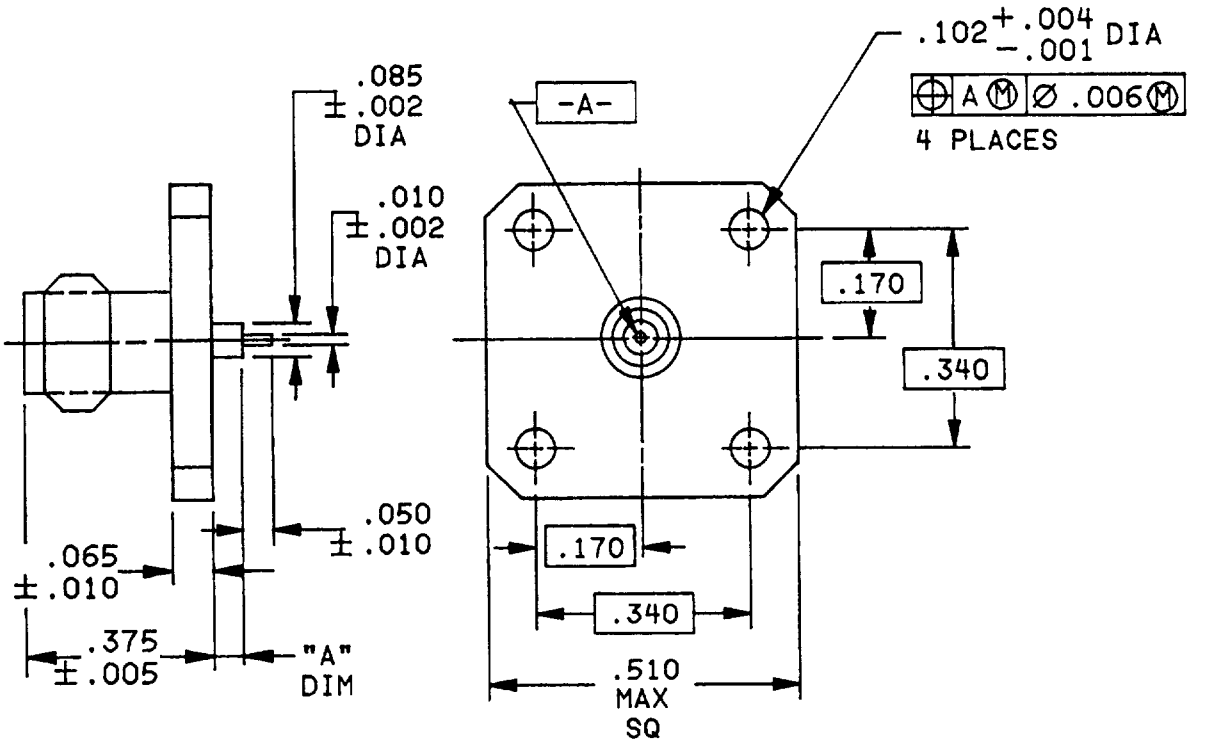
INCHES	MM	INCHES	MM
001	.03	.065	1 65
.002	.05	.085	2 16
.003	.08	.102	2.59
.004	.10	.112	2.84
.005	.13	.125	3.18
.006	.15	.223	5.66
.010	.25	2405	6.11
.050	1.27	.375	9.53
.057	1.45	.4810	12 22
		.625	15.88

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 1. Series SMA, socket contact, (2 hole) flange mounted receptacle, extended dielectric.

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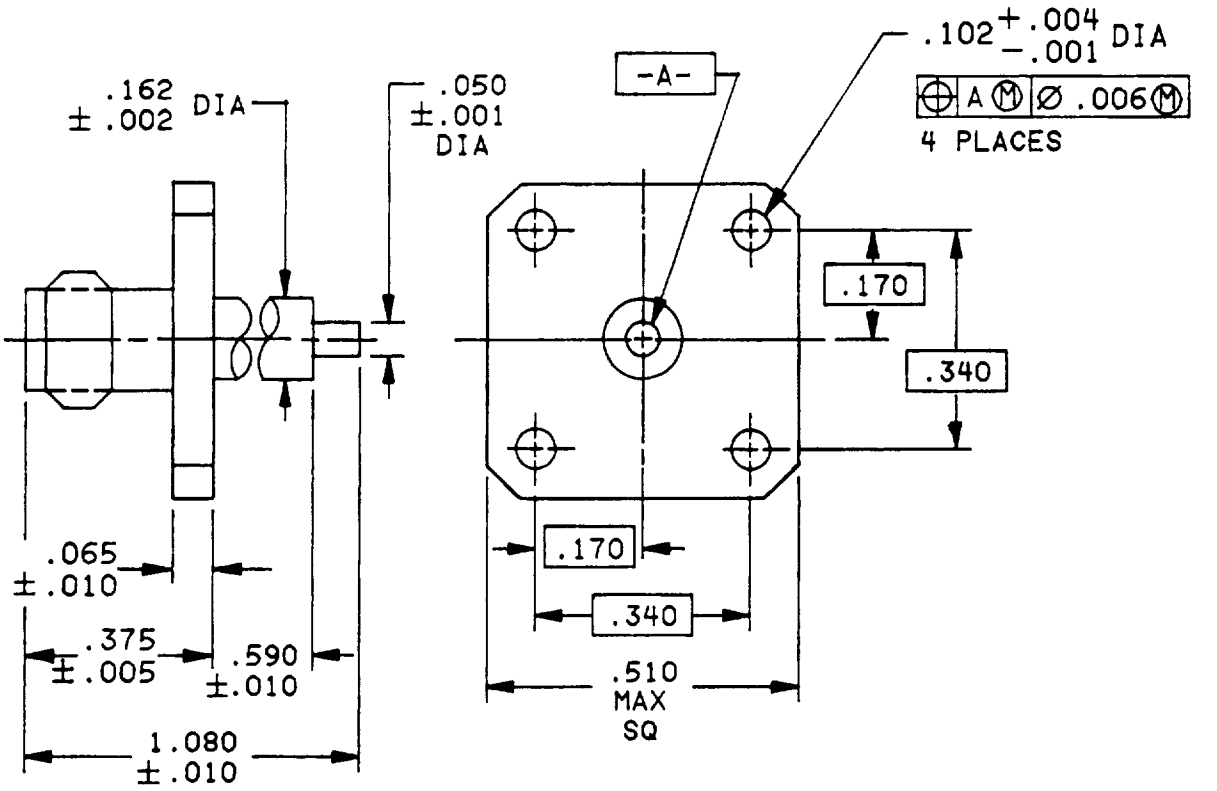


INCHES	MM	INCHES	MM
.001	.03	.065	1.65
.002	.05	.085	2.16
.004	.10	.102	2.59
.005	.13	.170	4.32
.006	.15	.340	8.64
.010	.25	.375	9.53
.010	.25	.510	12.95
.050	1.27		

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Number two screws are suggested for mounting.

FIGURE 2. Series SMA, socket contact, (4 hole) flange mounted receptacle .057 or .125 extended dielectric.

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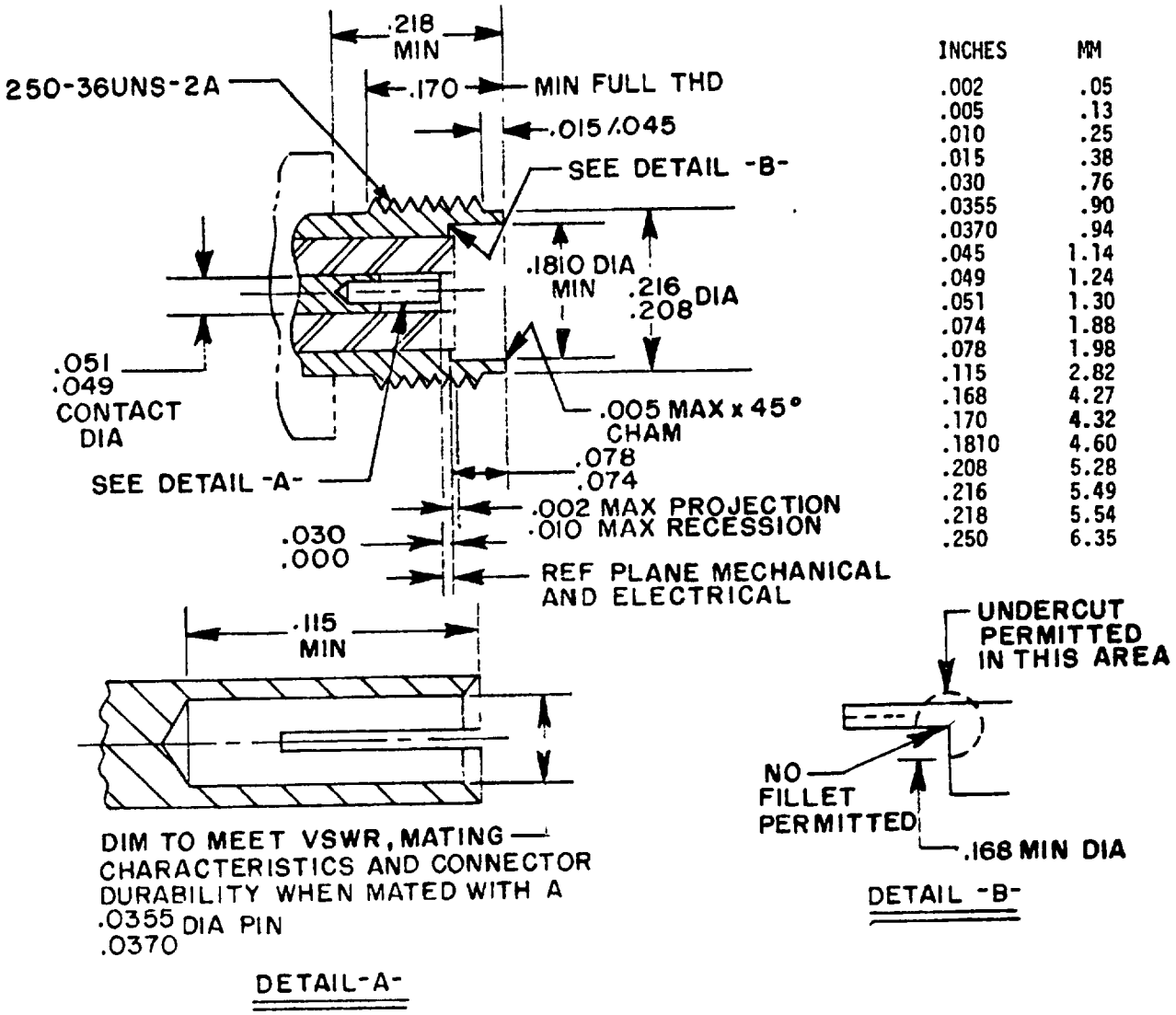


INCHES	MM	INCHES	MM
.001	.03	.102	2.59
.002	.05	.162	4.11
.004	.10	.170	4.32
005	.13	340	8.64
.006	.15	.375	9.53
.010	.25	.510	12.95
.050	1.27	.590	14.99
.065	1.65	1.080	27.43

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.
4 Number two screws are suggested for mounting.

FIGURE 3. Series SMA, socket contact, (4 nole) flange mounted receptacle, .590 (14.99 mm) extended dielectric.

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INCHES	MM
.002	.05
.005	.13
.010	.25
.015	.38
.030	.76
.0355	.90
.0370	.94
.045	1.14
.049	1.24
.051	1.30
.074	1.88
.078	1.98
.115	2.82
.168	4.27
.170	4.32
.1810	4.60
.208	5.28
.216	5.49
.218	5.54
.250	6.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 4. Mating dimensions for socket termination.

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ENGINEERING PARAMETERS:

Nominal impedance: 50 ohms.

Voltage rating: 600 Vrms maximum at sea level.
150 Vrms maximum at 70,000 feet.

Frequency range: 0 to 18.0 GHz.

Temperature rating: -65° to 105°C.

REQUIREMENTS:

Design and construction: See figures 1 through 4 and table I.

Force to engage and disengage:

Torque - 2 inch-pounds maximum.

Longitudinal force - Not applicable.

Coupling proof torque: Not applicable.

Inspection note: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Contact gaging: See figure 5.

Contacts with spring members:

Center contact (socket)

Oversize test pin - .0375 +.0001

Test pin finish - 16 microinches.

Insertion depth - .030/.045.

Number of insertions - 3.

Insertion force test: Steel test pin diameter .0370 +.0001.

Insertion depth - .050/.075.

Test pin finish - 16 microinches.

Insertion force - 3 pounds maximum.

Withdrawal force test: Steel test pin diameter .0355 - .0001.

Insertion depth - .050/.075.

Withdrawal force - 1 ounce minimum.

Test pin finish - 16 microinches.

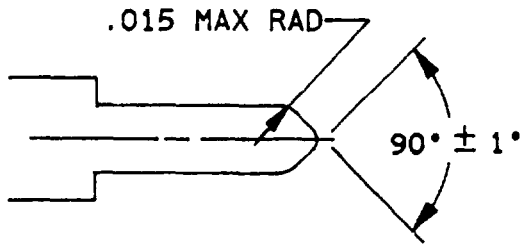


FIGURE 5. Test pin data.

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TABLE I. Part number and characteristics.

Part no. M83517/04-	Connector figure no.	Characteristics
31001	1	2 hole flange receptacle, .010 diameter terminal for .062 thick panel.
31002	1	2 hole flange receptacle, .010 diameter terminal for .125 thick panel.
31003	2	4 hole flange receptacle, .010 diameter terminal for .062 thick panel.
31004	2	4 hole flange receptacle, .010 diameter terminal for .125 thick panel.
31005	3	4 hole flange receptacle, .050 diameter terminal.

Permeability of nonmagnetic materials: Applicable.

Seal:

Hermetic sealed connectors: Not applicable.

Pressurized and weatherproof connectors: Not applicable.

Insulation resistance. 5,000 megohms minimum.

Center contact retention·

Axial force. 6 pounds minimum.

Torque. 4 inch-ounces minimum.

Dielectric withstanding voltage. Applicable, test condition I.

Test voltage 1,000 Vrms.

Corrosion. Applicable, test condition B.

Voltage standing wave ratio (VSWR).

Test frequency range From .5 to 18.0 GHz.

Swept frequency VSWR test setup

Step 1. See basic specification.

Step 2 VSWR shall be less than 1.080 +.005 frequency (frequency in GHz).

Item 11p VSWR shall be less than 1.025 + .002 frequency (frequency in GHz).

Item 11j VSWR shall be less than 1.025 + .002 frequency (frequency in GHz).

Step 3: VSWR shall be less than (to be determined).

Test fixture - See figure 1 of basic specification.

Step 4. VSWR test shall be less than (to be determined).

RF transmission loss (To be determined).

RF leakage Not applicable

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Connector durability.

Interface.

500 cycles minimum at 12 cycles/minute maximum rate.

Connector shall meet contact gaging and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact.	3.0	4.0
Outer contact.	2.0	Not applicable

Thermal shock: Applicable, test condition A.

Moisture resistance: Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

RF high potential withstanding voltage

At a frequency between 5 to 7.5 MHz.

Leakage current - Not applicable.

RF voltage - 1,000 Vrms.

Coupling mechanism retention force Not applicable.

Part number: M83517/4-(dash number from table 1.)

Group qualification: See table II.

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors 1/2/	Qualifies the following connectors
I	3†001	3†001 3†002
II	3†005	3†005

1/ Individual connectors other than listed are self qualifying only.

2/ Qualification of connectors qualifies connectors of the same material only.

† Denotes finish.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, AT
Navy - SH, AS
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 85

(Project 5935-3159-4)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-83517/4, Connector, Coaxial

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